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# Build the Geometry In, Don’t Hope It Emerges: A Reflection-Symmetry Audit of a Skeleton World Model

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## Abstract

1 The human body has an approximate left–right structure: reflect a pose horizon-  
2 tally and swap anatomical left and right landmarks, and any signed left-versus-  
3 right motion quantity should reverse sign. We use this group action to audit what  
4 a self-supervised skeleton model represents. The target is a coordinate-derived  
5 comparison of motion on the two sides, not a diagnosis or clinical gait measure.  
6 The quality-controlled cohort holds 625 pose sequences from 93 source videos,  
7 split at the source-video level so no clip or mirror from a test video reaches the  
8 matching training set, with a fresh encoder trained for each of fifty registered  
9 fold-and-seed combinations. Across every checkpoint the trained representation  
10 does not predict the laterality contrast better than an untrained encoder (learned  
11 minus untrained source-balanced  $R^2 = -0.02$ , 95% CI  $[-0.04, 0.00]$ ), it is not  
12 reflection-equivariant, and training moves it slightly *away* from the approximate  
13 equivariance at initialization ( $q$  rises by 0.03). Reflection augmentation does not  
14 repair this. Only an analytic odd/even read-out gives exact antisymmetry, and it  
15 adds no learned content, since the same construction on a random encoder does  
16 at least as well. For a skeleton world model, bilateral symmetry is something you  
17 install, not something you get for free from prediction.

## 18 1 Motivation and target

19 A human pose representation has an approximate left–right structure. Reflect a skeleton horizon-  
20 tally and exchange its anatomical left and right landmarks, and a signed left-versus-right motion  
21 quantity should reverse sign. This group action gives a direct handle on what a geometry-aware  
22 world model has captured about the articulated body. We use it to audit a self-supervised Skele-  
23 ton Joint-Embedding Predictive Architecture (S-JEPA) trained on the Gait Abnormality in Video  
24 Dataset (GAVD). The model predicts hidden latent content and never sees the laterality target or a  
25 dataset annotation during representation training.

26 For the left and right shoulders, knees, ankles, heels, and foot-index points, the target compares  
27 median speeds only on frame transitions where both sides are observed at both endpoints. For pair  
28  $k$ ,

$$c_k = \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + \epsilon}, \quad y = \text{mean}_k c_k.$$

29 All five pairs must each provide at least eight shared valid transitions. Mirroring negates  $y$ , changing  
30 invalid-coordinate placeholders leaves it unchanged, and mirroring twice restores the input; interpo-  
31 lated values may enter encoder input but never define the target. We keep three questions apart: a  
32 wrapper that forces sign change proves only the wrapper is correct; a trained encoder with a probe

33 showing sign-reversing behavior needs empirical support; and a direct token equivariance test is  
 34 stronger still. The audit measures all three.

## 35 2 Cohort and source-level split

36 The GAVD inventory contains 666 annotation files for 103 source videos [Ranjan et al., 2025, GAVD  
 37 project, 2026]. A matching pose archive is available for 642 annotations. Pre-specified quality-  
 38 control rules retain 625 pose sequences from 93 source videos and exclude 17; inclusion depends  
 39 on target computability and pose coverage, never on the sign or magnitude of the target. The split is  
 40 created at the source-video level because one source can yield several related sequences, and every  
 41 sequence, mirror, and augmentation inherits its source’s fold. GAVD supplies no persistent person  
 42 identifier, so this is held-out-video evaluation, not held-out-person evaluation.

Table 1: Five registered outer folds. Splitting and balancing operate on source videos.

| Fold | Train sources | Test sources | Train sequences | Test sequences |
|------|---------------|--------------|-----------------|----------------|
| 0    | 74            | 19           | 436             | 189            |
| 1    | 74            | 19           | 443             | 182            |
| 2    | 74            | 19           | 553             | 72             |
| 3    | 75            | 18           | 548             | 77             |
| 4    | 75            | 18           | 520             | 105            |

## 43 3 Training, validation, and evaluation

44 A fresh encoder is trained for every outer fold, seed, and variant: five folds, five optimization seeds  
 45 (42–46), and two variants—vanilla and reflection-augmented—for 50 fold-local encoders, each fol-  
 46 lowing a fixed 300-epoch schedule on its outer-training sources with no early-stopping validation set.  
 47 Within a seed the variants share initialization, source draws, masks, views, and optimizer schedule,  
 48 with reflection on a separate random stream. Validation is reserved for the linear ridge read-out: four  
 49 inner folds inside each outer-training set select the ridge penalty and nothing else, after which the  
 50 read-out is refitted on all outer-training sources and applied once to the untouched outer-test sources,  
 51 with fitted preprocessing estimated from outer-training data only.

52 The primary measure is source-balanced  $R^2$ . Comparison lanes include a paired untrained encoder, a  
 53 target-component self-consistency oracle (a verification, not a learned baseline), measured nuisance  
 54 features, analytically antisymmetric odd/even read-outs, and a direct encoder-token reflection test  
 55 that compares tokens under the full anatomical permutation with no channel fitting, rotation, sign  
 56 search, or read-out; zero is exact equivariance. Five seeds describe optimization variation, and 2,000  
 57 source-video bootstrap resamples—keeping each video’s sequences and seeds together—describe  
 58 uncertainty over the observed collection, conditional on the fitted pipeline.

## 59 4 Results

60 The registered run supports none of the emergence claims and passes only the construction’s own  
 61 integrity check. The pattern holds for the reflection-augmented variant and for a high-coverage  
 62 sensitivity subset (600 sequences from 91 sources at  $\geq 0.9$  coverage), where the primary  $R^2$  is  
 63 again near 0.06.

- 64 • **Predictive utility does not clear the floor.** The learned free read-out reaches  $R^2 = 0.06$   
 65 (95% CI  $[-0.03, 0.13]$ ); against the untrained encoder, training changes held-out  $R^2$  by  
 66  $-0.02$  (95% CI  $[-0.04, 0.00]$ ). The oracle recovers the target almost perfectly ( $R^2 \approx 1.00$ ),  
 67 so the failure is in the learned features. Measured nuisances add nothing.
- 68 • **Reflection augmentation does not help:** the augmented encoder differs from vanilla by  
 69  $0.00$  (95% CI  $[-0.01, 0.01]$ ) on the primary lane.
- 70 • **The construction works, but the encoder is not why.** The odd/zero-origin read-out is  
 71 exactly antisymmetric for every seed, yet weak ( $R^2 = 0.04$ , 95% CI  $[-0.04, 0.11]$ ), and the

same construction on an untrained encoder does at least as well (learned minus untrained  $-0.06$ , 95% CI  $[-0.10, -0.02]$ ). Training nudges only the even part ( $+0.02$ , 95% CI  $[0.01, 0.04]$ ).

- **Training degrades reflection equivariance.** The learned token error is  $q = 0.11$  (95% CI  $[0.10, 0.14]$ ), above the 0.10 margin, while the untrained initialization is closer at  $q = 0.08$  (95% CI  $[0.07, 0.09]$ ); learned minus initial is  $+0.03$  (95% CI  $[0.02, 0.05]$ ). The native free read-out is far from antisymmetric (output error 0.22, 95% CI  $[0.19, 0.24]$ ).

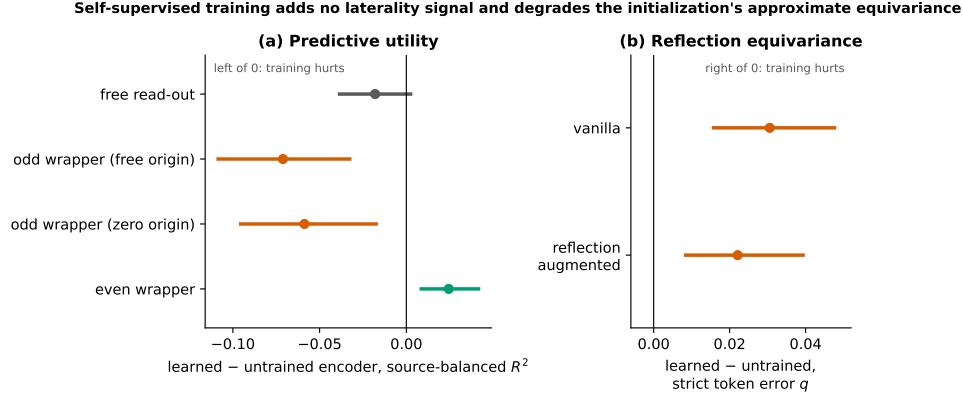


Figure 1: Learned minus untrained differences with 95% source-cluster bootstrap intervals. (a) No read-out shows a predictive gain from training. (b) The strict token error rises after training for both variants, so training moves the representation away from the equivariance present at initialization.

## 5 Claim boundary, ethics, and status

The strongest supported claim is post-development, within-GAVD cross-validated performance on held-out source videos. The protocol does not support diagnosis, clinical validity, prevalence, treatment effects, or generalization to unidentified people in new videos, and the null is specific to the registered architecture, seeds, target, and controls. Folder names are dataset annotations, not diagnoses. Earlier transductive estimates are superseded. Submission and artifact release stay blocked until the institutional ethics determination, the data-use review, and the derived-pose release review each carry a genuine dated internal reference. No raw video or identity-bearing frame is redistributed, and derived artifacts remain non-public unless completed reviews permit release.

## References

- GAVD project. Gait abnormality video dataset: Data access and responsible-use notes. GitHub repository, 2026. URL <https://github.com/Rahmyyy/GAVD>. Accessed: 2026-09-03.
- Rahm Ranjan, David Ahmed-Aristizabal, Mohammad Ali Armin, and Juno Kim. Computer vision for clinical gait analysis: A gait abnormality video dataset. *IEEE Access*, 13:45321–45339, 2025. doi: 10.1109/ACCESS.2025.3545787.